



DairyDesk

Complete Setup Guide — From Zero to Running the App

Windows 10 / 11

PostgreSQL 16

Django + DRF

Vite + React

Docker

For a fresh Windows PC with **none** of the required software installed.
Repository: github.com/TanmayKamble004/DairyDesk · Prepared July 2026

Who this is for. You have a Windows laptop with nothing installed yet. Follow the sections **top to bottom** — don't skip ahead. Part A installs the four tools the project needs; Part B gets the code and runs it. Total time: roughly **30–45 minutes**, most of it downloads.

What you'll end up with

Piece	What it is	Runs at
Database	PostgreSQL 16 inside Docker	<code>localhost:5432</code>
Backend	Django REST API (the server)	<code>localhost:8000</code>
Frontend	React app (what you look at)	<code>localhost:5173</code>

Part A — Install the required software

The project needs **four** tools. Install them in this order. After each one, **close and reopen any terminal** so it picks up the new install, then run the verify command.

How to open a terminal (PowerShell): Press **Windows key**, type **PowerShell**, and click **Windows PowerShell**. You'll use this a lot.

1 Git — downloads the code

Download and run the installer from <https://git-scm.com/downloads> (choose **64-bit Git for Windows Setup**).

- ☐ Keep clicking **Next** — the default options are all fine.
- ☐ This also installs **Git Bash**, an alternative terminal (optional).

Verify (in a fresh PowerShell window):

```
git --version
```

You should see something like `git version 2.4x.x`.

2 Python 3.12+ — runs the backend

Download from <https://www.python.org/downloads/> (the big yellow **Download Python 3.12.x** button).

Critical: On the **first** installer screen, tick the box “**Add python.exe to PATH**” at the bottom before clicking **Install Now**. If you miss it, the commands later won't work.

Verify:

```
python --version
```

You should see `Python 3.12.x` (or newer).

3 Node.js 20 LTS — runs the frontend

Download the **LTS** version from <https://nodejs.org/> (the left-hand button that says “**LTS**”). Run the installer and click **Next** through the defaults; the default options are fine.

Verify:

```
node --version  
npm --version
```

`node` should be `v20.x` or newer, and `npm` should print a version too.

4 Docker Desktop — runs the database

Download from <https://www.docker.com/products/docker-desktop/> and run the installer with default options.

Restart your PC if it asks you to.

- ☐ After installing, **launch Docker Desktop** from the Start menu.
- ☐ Wait until the whale icon in the taskbar (bottom-right) stops animating and the app says “**Engine running**”. First launch can take a minute or two.
- ☐ If prompted, accept the service agreement and skip / close the sign-in — an account is not required.

If Docker complains about WSL 2: it will show a link to install the WSL 2 update — or just open PowerShell **as Administrator** and run `wsl --install`, then restart. Reopen Docker Desktop afterwards.

Verify (with Docker Desktop open and running):

```
docker --version  
docker compose version
```

Checkpoint. All four should print a version with no errors: `git`, `python`, `node` + `npm`, and `docker`. If any fail, close every terminal, reopen a fresh one, and try again — new installs only show up in newly opened terminals.

Part B — Get the code and run it

1 Get access & clone the repository

DairyDesk is a **private** repository, so Tanmay must add you as a collaborator on GitHub first (you'll get an email invite — accept it). Then, in PowerShell, pick a folder for the project and clone it:

```
# Go to your Documents folder (or wherever you want the project)
cd $HOME\Documents

# Download the code
git clone https://github.com/TanmayKamble004/DairyDesk.git
cd DairyDesk
```

Sign-in popup: the first time you clone, a browser window or dialog asks you to sign in to GitHub. Sign in with the account that received the invite. If nothing pops up and it fails, install [GitHub CLI](#) and run `gh auth login` once.

2 Create the environment files

The app reads its settings from `.env` files. Copy the provided templates — the default values work for local development as-is:

```
Copy-Item .env.example .env
Copy-Item frontend\.env.example frontend\.env
```

3 Start the database (Docker)

Make sure **Docker Desktop is open and running** (whale icon steady). Then, from the `DairyDesk` folder:

```
docker compose up -d
```

This downloads and starts PostgreSQL 16 in the background. Confirm it's healthy:

```
docker compose ps
```

Wait until the `db` row shows **(healthy)** — a few seconds after first start.

4 Set up and start the backend (Django)

Still in the `DairyDesk` folder, move into the backend and create an isolated Python environment (a “virtual environment”), then activate it:

```
cd backend
python -m venv .venv
.\.venv\Scripts\Activate.ps1
```

If PowerShell blocks the activate script ("running scripts is disabled"), run this once, then re-run the activate line above:

```
Set-ExecutionPolicy -Scope CurrentUser RemoteSigned
```

When activated, your prompt shows `(.venv)` at the start of the line.

Now install the backend's Python packages, set up the database tables, load demo data, and start the server:

```
pip install -r requirements.txt
python manage.py migrate
python manage.py seed_demo
python manage.py runserver
```

When you see **"Starting development server at http://127.0.0.1:8000/"**, the backend is up.

Leave this terminal running. Closing it stops the backend. You'll open a **second** terminal for the frontend in the next step.

5 Set up and start the frontend (React)

Open a **new** PowerShell window (leave the backend one running). Go to the project's frontend folder and start it:

```
cd $HOME\Documents\DaairyDesk\frontend
npm install
npm run dev
```

`npm install` downloads the frontend packages (takes a minute the first time). When you see a **Local: http://localhost:5173/** line, it's ready.

6 Open the app & log in

In your browser, go to <http://localhost:5173>. Log in with one of the demo accounts created by `seed_demo`:

Role	Username	Password	Access
Owner	<code>owner</code>	<code>owner123</code>	Full access + Django admin
Staff	<code>staff</code>	<code>staff123</code>	Limited access

 **That's it — DairyDesk is running.** Explore the 3D inventory shelf, products, customers, orders and invoices. `seed_demo` is safe to re-run anytime to reset the demo data.

Running it again next time

Once everything is installed, you **don't** repeat Part A. To start the app on any later day:

#	Do this	Where
1	Open Docker Desktop and wait for "Engine running"	Start menu
2	<code>docker compose up -d</code>	PowerShell in <code>DairyDesk</code>
3	<code>cd backend</code> → <code>.\.venv\Scripts\Activate.ps1</code> → <code>python manage.py runserver</code>	Terminal 1
4	<code>cd frontend</code> → <code>npm run dev</code>	Terminal 2 (new)
5	Open http://localhost:5173	Browser

To stop everything: press `Ctrl + C` in both terminals, then run `docker compose down` to stop the database. Your data is preserved for next time.

Troubleshooting

Symptom	Fix
"Docker daemon is not running" / <code>docker</code> errors	Open Docker Desktop and wait for the whale icon to go steady ("Engine running"), then retry.
Backend: database connection refused	The DB wasn't ready. Run <code>docker compose ps</code> and wait for (healthy) , then start the backend again.
PowerShell won't activate the venv ("scripts is disabled")	Run once: <code>Set-ExecutionPolicy -Scope CurrentUser RemoteSigned</code> , then re-run the activate line.
" <code>python</code> / <code>node</code> / <code>git</code> is not recognized"	Close every terminal and open a fresh one. If still failing, reinstall that tool — for Python, make sure "Add to PATH" was ticked.
"Port already in use" (8000, 5173, or 5432)	Another program is using that port. Close it, or restart your PC and try again.
<code>npm install</code> fails partway	Delete the <code>frontend\node_modules</code> folder and run <code>npm install</code> again on a stable connection.
Login fails / no demo accounts	In the backend terminal (venv active), re-run <code>python manage.py seed_demo</code> .